

Mutual Fund Analytics Capstone Project

Bluestock Fintech Internship

Project Title: Mutual Fund Analytics &
Interactive Dashboard Development

Prepared By:
Divyansh Dave

Internship Program: Bluestock Fintech

Project Duration: June 2026

Technology Stack:
Python • Pandas • NumPy • SQLite • SQL •
Power BI • Git & GitHub

Final Project Report

Executive Summary

The Mutual Fund Analytics Capstone Project was developed as part of the Bluestock Fintech Internship to demonstrate an end-to-end data analytics workflow. The objective of the project was to collect, clean, analyze, and visualize mutual fund and investor data to generate meaningful business insights.

The project involved building a complete ETL (Extract, Transform, Load) pipeline using Python and Pandas to process multiple datasets, including mutual fund details, NAV history, transaction records, and fund performance metrics. The cleaned data was stored and managed efficiently for further analysis.

Exploratory Data Analysis (EDA) was performed to understand investor behavior, transaction patterns, and fund performance. Advanced analytics techniques such as Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio analysis, Investor Cohort Analysis, SIP Continuity Analysis, and a simple fund recommendation engine were implemented to provide deeper insights into investment risk and performance.

An interactive four-page Power BI dashboard was designed to present key metrics, trends, and visualizations in a user-friendly format. The dashboard enables users to explore fund performance, investor demographics, transaction distributions, and risk-return relationships.

Overall, this project demonstrates practical skills in data engineering, data analysis, business intelligence, and visualization while providing a structured framework for mutual fund performance evaluation and investor analytics.

1. Problem Statement

The mutual fund industry produces large volumes of data related to NAV, fund performance, investor transactions, and SIP activities. Since this data is spread across multiple sources, it becomes difficult to analyze performance, assess risks, and derive meaningful insights. This project aims to integrate, clean, analyze, and visualize mutual fund data through an interactive analytics platform and dashboard.

2. Project Objectives

- Build an end-to-end ETL pipeline for mutual fund data.
 - Clean and preprocess datasets using Python and Pandas.
 - Store processed data in SQLite.
 - Perform Exploratory Data Analysis (EDA).
 - Calculate key metrics such as returns, Sharpe Ratio, VaR, and CVaR.
 - Analyze investor transactions and SIP continuity.
 - Develop a 4-page Power BI dashboard.
 - Generate advanced analytics and a simple fund recommendation system.
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3. Project Scope

In Scope

- Mutual fund and NAV analysis.
- Investor transaction and SIP analysis.
- Risk and performance measurement.
- Interactive dashboards and analytics reports.

Out of Scope

- Real-time market prediction.
- Live trading recommendations.
- Automated investment execution.

4. Data Sources & Dataset Description

The Mutual Fund Analytics Capstone Project utilizes multiple datasets containing information related to mutual fund schemes, historical NAV values, investor transactions, and fund performance. These datasets were cleaned, transformed, and integrated to perform comprehensive analytics and build interactive dashboards.

Datasets Used

Dataset Name	Description
fund_master.csv	Contains master information about mutual fund schemes, including scheme names, AMFI codes, categories, and fund houses.
nav_history.csv	Stores historical Net Asset Value (NAV) data for each mutual fund scheme across different dates.
investor_transactions.csv	Includes investor transaction records such as SIPs, lump sum investments, redemptions, amounts, and transaction dates.
performance.csv	Contains calculated performance metrics such as returns, CAGR, Sharpe Ratio, volatility, and risk grades.

Data Processing

The raw datasets underwent several preprocessing steps before analysis:

- Standardization of date formats and column names.
- Removal of duplicate records.
- Handling of missing and invalid values.
- Validation of transaction amounts and NAV values.
- Feature engineering for performance and risk metrics.

After preprocessing, the cleaned datasets were stored in the **data/processed/** directory and used for Exploratory Data Analysis (EDA), advanced analytics, and Power BI dashboard creation.

Processed Files Used in the Project

- clean_fund_master.csv
- clean_nav_history.csv

- clean_transactions.csv
- clean_performance.csv

5. ETL Pipeline & Data Cleaning Process

5.1 Overview

The Mutual Fund Analytics project follows a structured **ETL (Extract, Transform, Load)** process to convert raw data into a clean and analysis-ready format. The ETL pipeline was developed using **Python, Pandas**, and **SQLite**, ensuring consistency and reliability across all datasets.

5.2 Extract (Data Collection)

The project began by importing multiple raw CSV datasets containing information about:

- Mutual fund master details.
- Historical NAV (Net Asset Value) records.
- Investor transactions.
- Fund performance metrics.

These datasets were loaded into Python using the Pandas library for further processing.

5.3 Transform (Data Cleaning & Preprocessing)

Several preprocessing operations were performed to improve data quality:

- Removed duplicate records.
- Standardized date formats using datetime.
- Handled missing and invalid values.
- Verified that NAV and transaction amounts were greater than zero.
- Standardized transaction types (SIP, Lumpsum, Redemption).
- Validated KYC status values.
- Created derived metrics and cleaned column names.

The cleaned datasets were then stored as:

- clean_fund_master.csv
- clean_nav_history.csv
- clean_transactions.csv

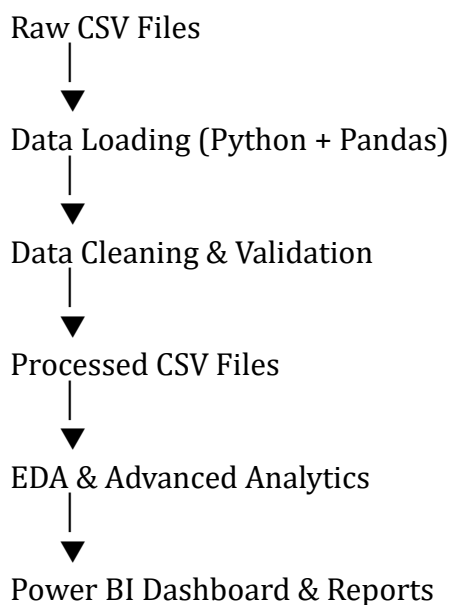
- clean_performance.csv

5.4 Load (Processed Data Storage)

After cleaning and transformation, the processed datasets were saved in the **data/processed/** folder. These cleaned files served as the primary data source for:

- Exploratory Data Analysis (EDA),
- Advanced Analytics,
- Power BI Dashboard development,
- Risk analysis and reporting.

5.5 ETL Workflow Diagram

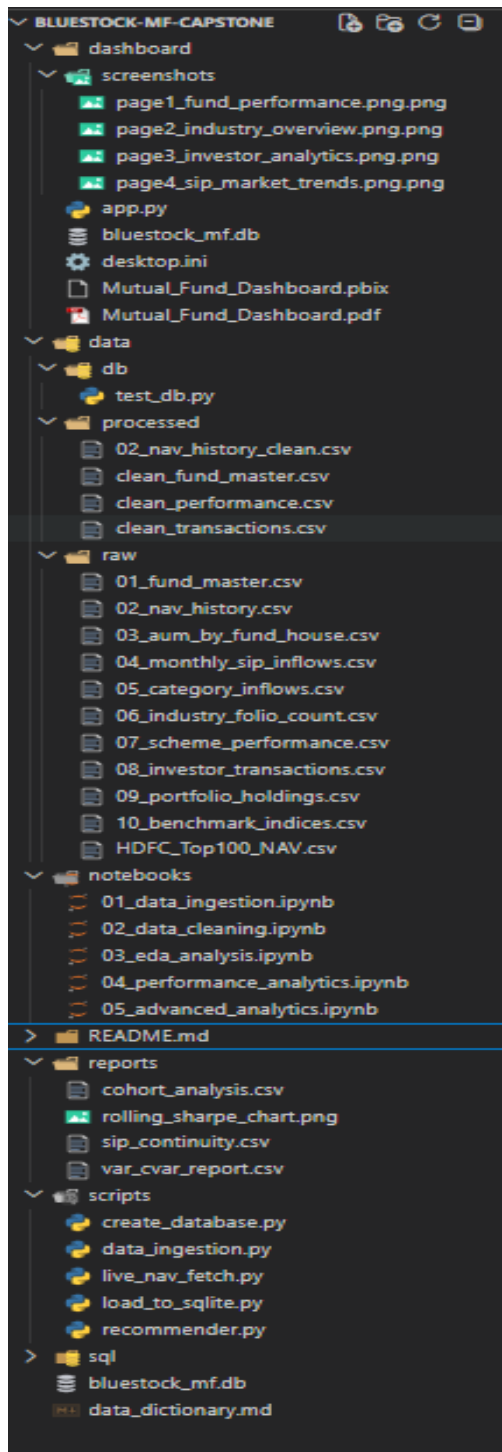


5.6 Technologies Used

Tool / Technology Purpose

Python	Data processing and analytics
Pandas	Data cleaning and transformation
NumPy	Numerical computations
SQLite	Data storage and querying
Power BI	Dashboard creation and visualization
Git & GitHub	Version control and project management

Figure 1: Organized Project Repository Structure for the Mutual Fund Analytics Capstone Project



The project repository is organized into separate modules for raw and processed data, Jupyter notebooks, Python scripts, reports, and Power BI dashboard assets. This modular structure ensures better maintainability, reproducibility, and efficient project management throughout the data analytics lifecycle.

6. Exploratory Data Analysis (EDA)

6.1 Overview

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the mutual fund datasets and identify meaningful patterns. The analysis focused on fund categories, investor behavior, transaction trends, NAV movement, and fund performance metrics. Python libraries such as **Pandas**, **Matplotlib**, and **Seaborn** were used for data exploration and visualization.

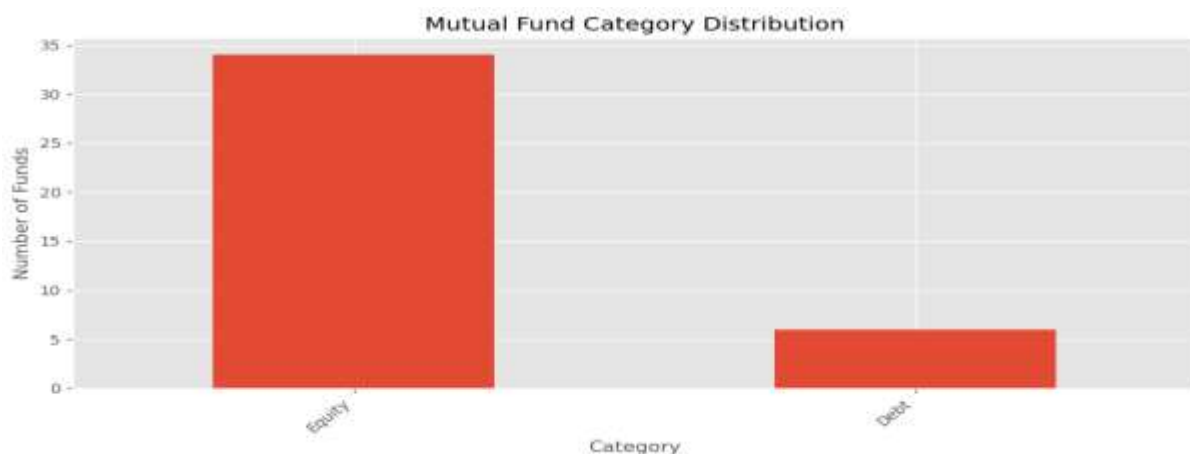
6.2 Key EDA Objectives

- Analyze the distribution of mutual fund categories.
- Study investor transaction patterns.
- Identify top-performing and high-risk funds.
- Examine monthly SIP inflows and trends.
- Explore the relationship between returns and risk grades.

6.3 Major Insights

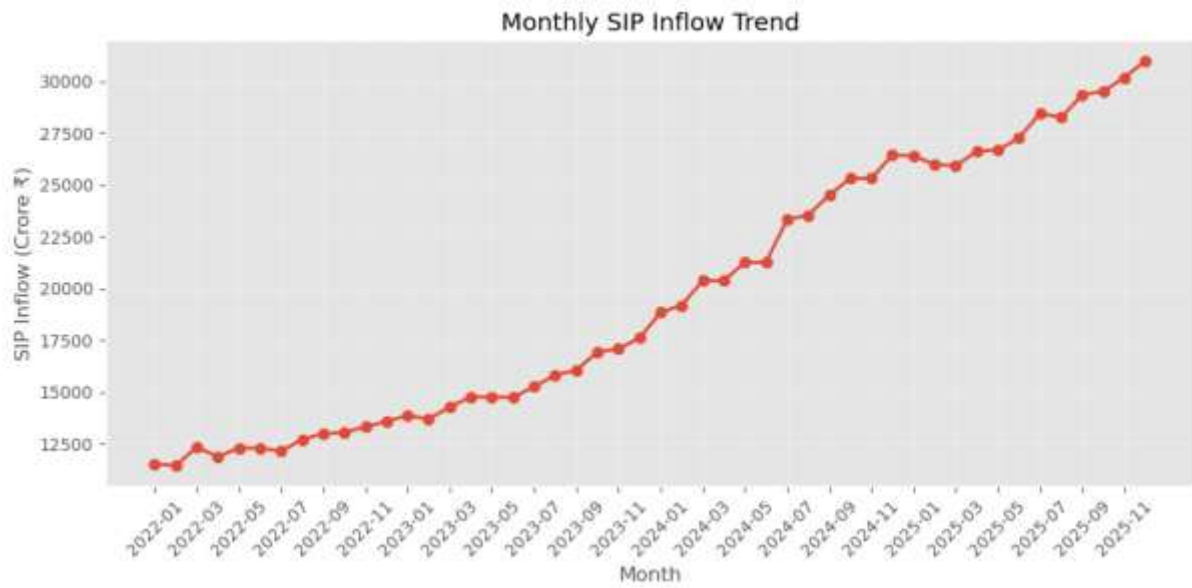
1. Mutual Fund Category Distribution

The dataset contains multiple mutual fund categories, with equity-oriented schemes representing a significant portion. Category analysis helps in understanding market segmentation and investor preferences.



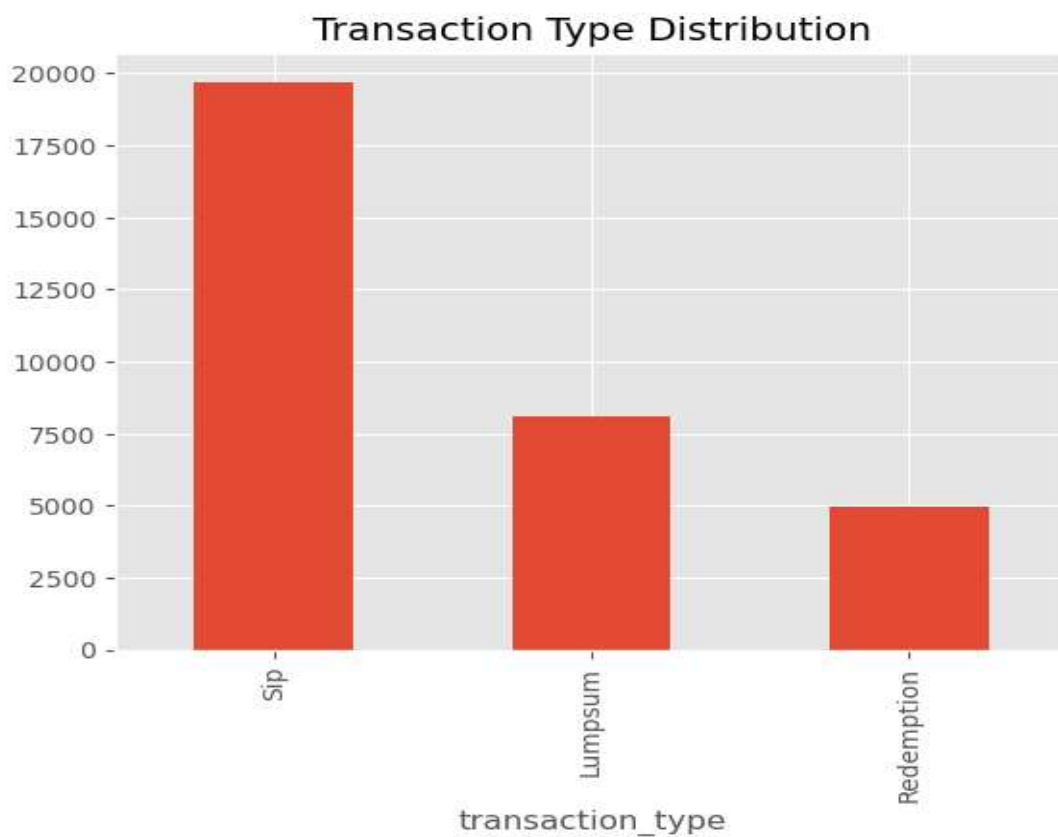
2. Monthly SIP Inflow Trends

Analysis of monthly SIP contributions highlights investment patterns over time. Stable and increasing SIP inflows indicate growing investor confidence in mutual fund investments.



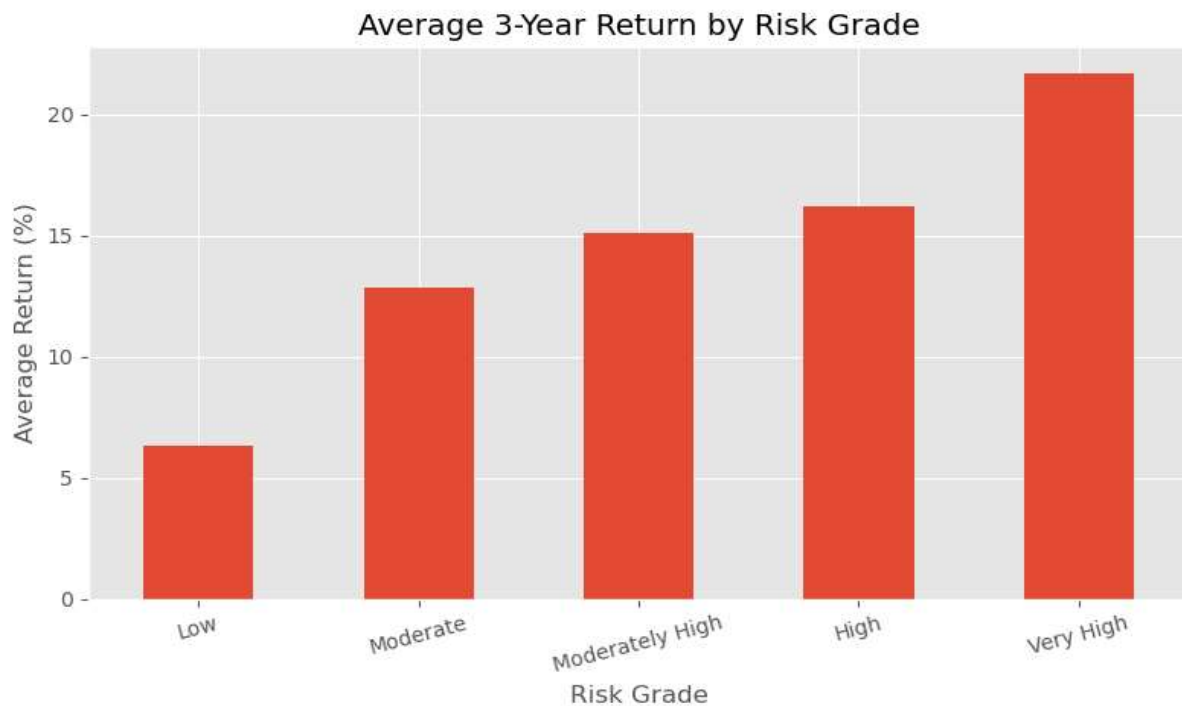
3. Investor Transaction Analysis

Investor transaction records were analyzed to compare SIPs, lump-sum investments, and redemption activities. This provides insights into investor behavior and fund liquidity.



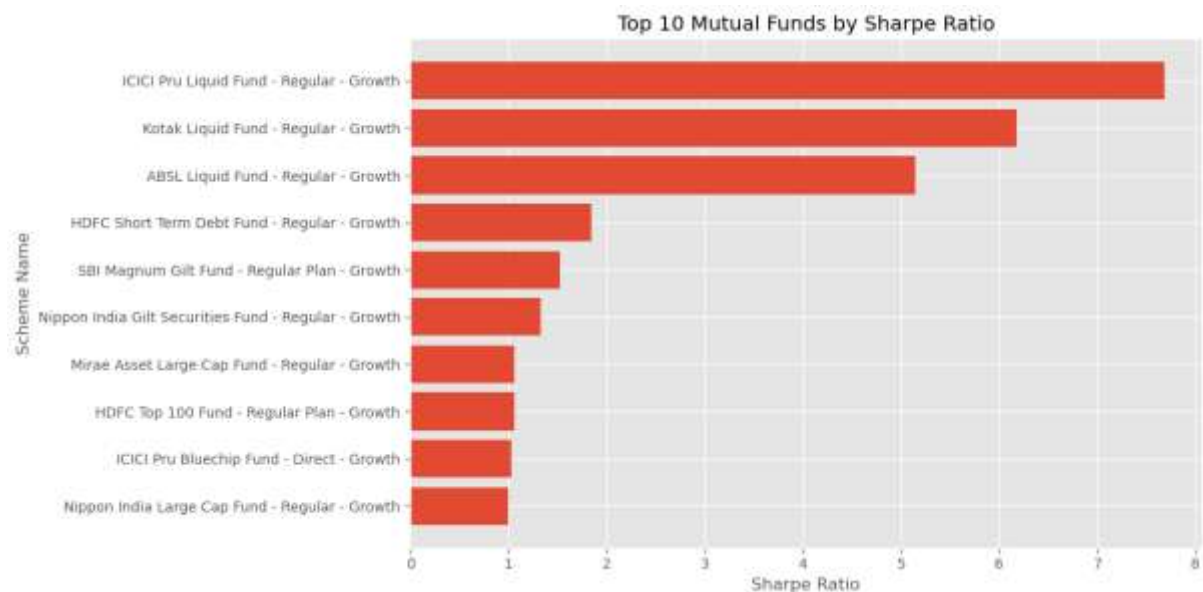
4. Risk Grade vs Average Return

Comparing average returns across different risk grades shows that higher-risk funds generally provide better long-term returns, although with increased volatility.



5. Top Performing Mutual Funds

Performance metrics such as Sharpe Ratio and multi-year returns were used to identify the best-performing schemes in the dataset.



6.4 Summary of EDA Findings

The EDA process revealed several important trends:

- Equity mutual funds dominate the dataset.
- SIP investments form a significant share of investor transactions.
- Higher-risk funds generally achieve higher average returns.
- Transaction and NAV data exhibit consistent patterns suitable for advanced risk analysis.
- The insights obtained during EDA formed the basis for the advanced analytics models and the Power BI dashboard.

7. Performance Analytics & Advanced Analytics

7.1 Overview

The final phase of the Mutual Fund Analytics project focused on advanced risk measurement and investor behavior analysis. Statistical and financial analytics techniques were implemented using Python to evaluate mutual fund performance, identify investor trends, and build a simple recommendation engine.

7.2 Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Historical Value at Risk (VaR) at the 95% confidence level was calculated using the daily NAV return distribution for each mutual fund scheme. Conditional Value at Risk (CVaR) was also computed to estimate the average loss beyond the VaR threshold. These metrics help quantify downside investment risk and support better portfolio evaluation.

Output Generated:

- reports/var_cvar_report.csv

Key Insight:

- Funds with higher expected returns generally exhibited larger VaR and CVaR values, indicating greater downside risk exposure.
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7.3 Rolling 90-Day Sharpe Ratio Analysis

A rolling 90-day Sharpe Ratio was computed using rolling average returns and rolling standard deviation. The analysis provides a dynamic view of risk-adjusted performance over time rather than relying on a single static value.

Output Generated:

- reports/rolling_sharpe_chart.png

Key Insight:

- Rolling Sharpe analysis revealed periods of improved and reduced risk-adjusted performance, helping identify stable and volatile market phases.



7.4 Investor Cohort Analysis

Investor transactions were grouped according to the year of their first investment. For each cohort, average SIP amount, total investment value, and preferred mutual fund categories were analyzed.

Output Generated:

- reports/cohort_analysis.csv

Key Insight:

- Cohort analysis highlights long-term investor engagement patterns and helps identify which investor groups contribute most significantly to assets under management.

7.5 SIP Continuity Analysis

A SIP continuity study was performed to identify gaps between consecutive SIP transactions. Investors with gaps greater than 35 days were flagged as "at-risk," allowing early detection of potential SIP discontinuation.

Output Generated:

- reports/sip_continuity.csv

Key Insight:

- Monitoring SIP continuity enables proactive identification of investors who may require engagement or retention strategies.
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7.6 Mutual Fund Recommendation System

A simple recommendation engine was developed to suggest mutual funds based on the investor's selected risk appetite (Low, Moderate, or High). The recommender filters schemes by risk grade and ranks them using the Sharpe Ratio, returning the top three funds.

Script Implemented:

- scripts/recommender.py

Sample Recommendation Output:

Scheme Name	Risk Grade	Sharpe Ratio
Kotak Emerging Equity Fund - Regular - Growth	High	0.96
ICICI Pru Midcap Fund - Regular - Growth	High	0.95
DSP Midcap Fund - Regular - Growth	High	0.90

Key Insight:

- The recommendation engine combines risk preference with historical risk-adjusted returns, providing a simple but effective personalized fund selection approach.
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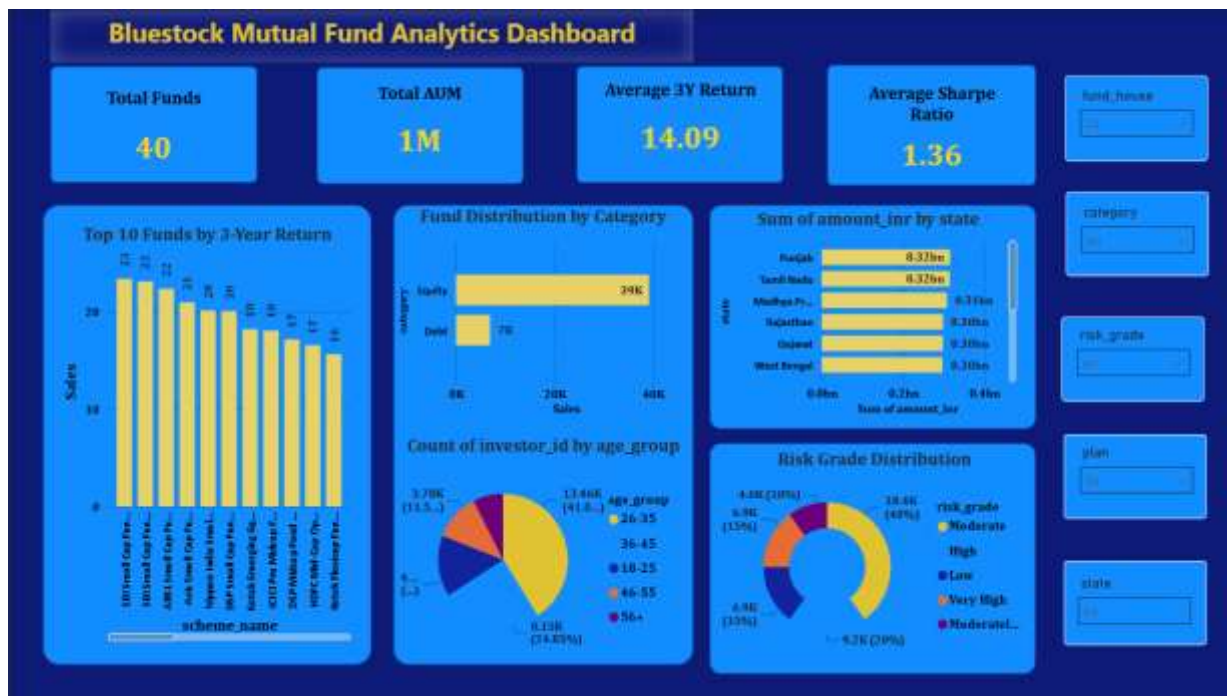
8. Power BI Dashboard Overview

An interactive 4-page Power BI dashboard was developed to present key insights from the mutual fund datasets. The dashboard integrates fund performance metrics, investor transaction patterns, SIP market trends, and category-wise analysis into a single visual platform. Interactive filters and charts enable users to explore data dynamically and support better investment analysis and decision-making.

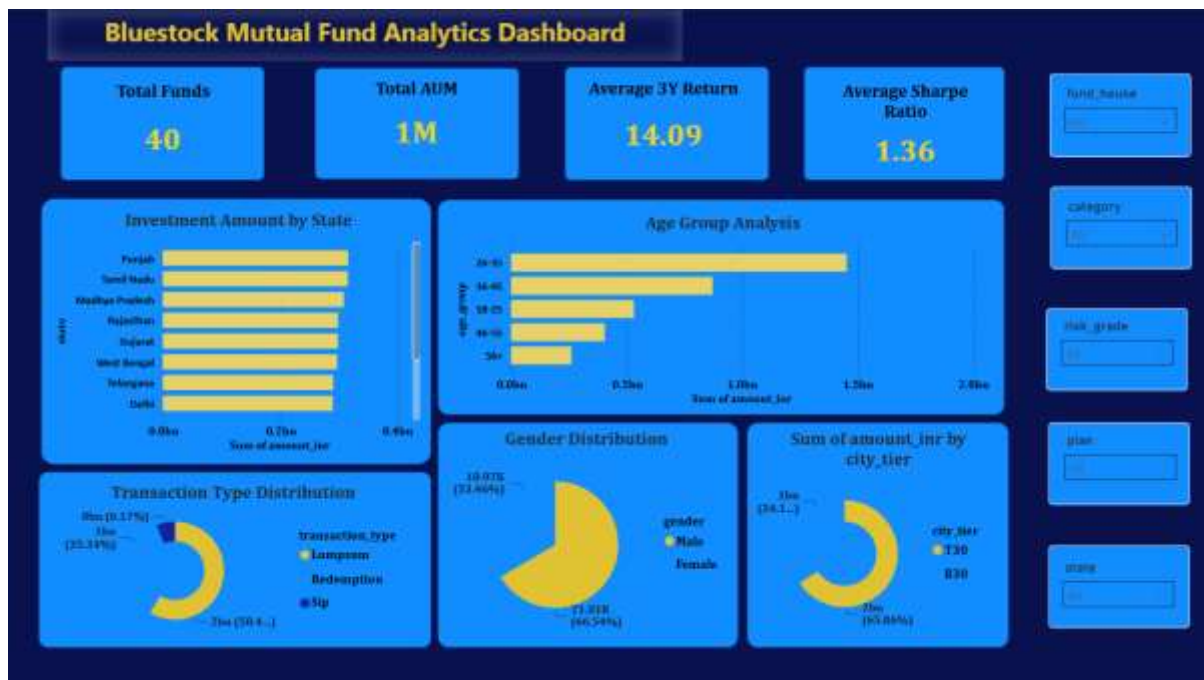
Dashboard Overview



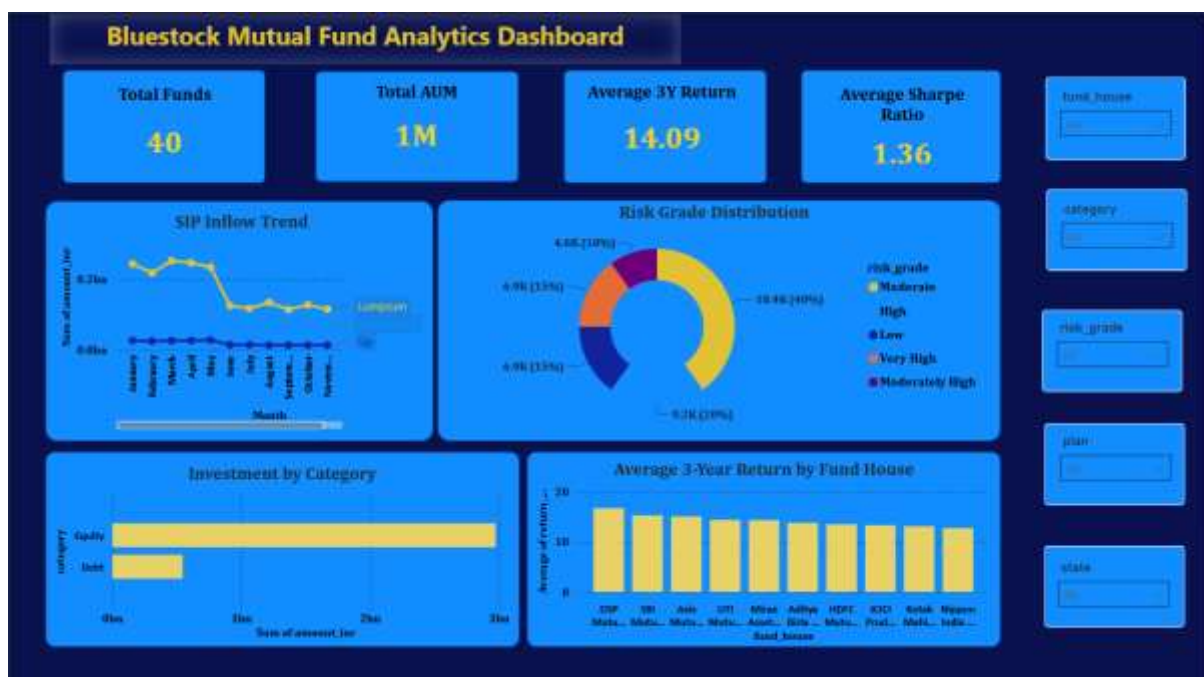
Category & Industry Analysis



Investor Analytics



SIP Market Trends



9. Conclusion and Future Scope

9.1 Conclusion

The Mutual Fund Analytics Capstone Project successfully integrated data engineering, exploratory analysis, advanced analytics, and interactive dashboard development into a single end-to-end solution. Using Python, SQL, Power BI, and GitHub, the project transformed raw mutual fund and investor datasets into actionable business insights.

The developed system provides:

- Comprehensive ETL and data cleaning pipeline.
- Exploratory analysis of mutual fund and investor behavior.
- Risk and performance metrics including Sharpe Ratio, VaR, and CVaR.
- Investor cohort and SIP continuity analysis.
- A Power BI dashboard for interactive data visualization.
- A basic mutual fund recommendation engine based on risk appetite.

This project demonstrates practical applications of data analytics and business intelligence techniques in the financial services domain.

9.2 Future Scope

Potential future enhancements include:

- Integration of live mutual fund NAV data through APIs.
 - Machine learning models for return prediction and fund ranking.
 - Personalized recommendation systems using collaborative filtering.
 - Real-time dashboard deployment using Power BI Service or Streamlit Cloud.
 - Automated data refresh and report generation pipelines.
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10. References

- Association of Mutual Funds in India (AMFI)
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Power BI Documentation
- SQLite Documentation
- Git & GitHub Documentation